

LAB PROGRESS REPORT · 1 OCTOBER 2026 · VERSION 2

Reading the Current

What Geneformer's "virtual gene edits" can tell us,
and the checks we needed before believing them

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Our question: which genes, if changed, would push a tumour's T cells back toward normal?

T cell inside
a lung tumour

change one gene



in the computer

T cell in the same
patient's normal lung

T cells are the immune cells that can kill tumour cells. Inside a tumour they often stop working well. If we knew which genes hold them in that state, we would know where to look for treatments. Testing every gene in the lab is slow and expensive, so we asked a computer model first.

The short answer: the model always answers, so we built checks to tell when it means something

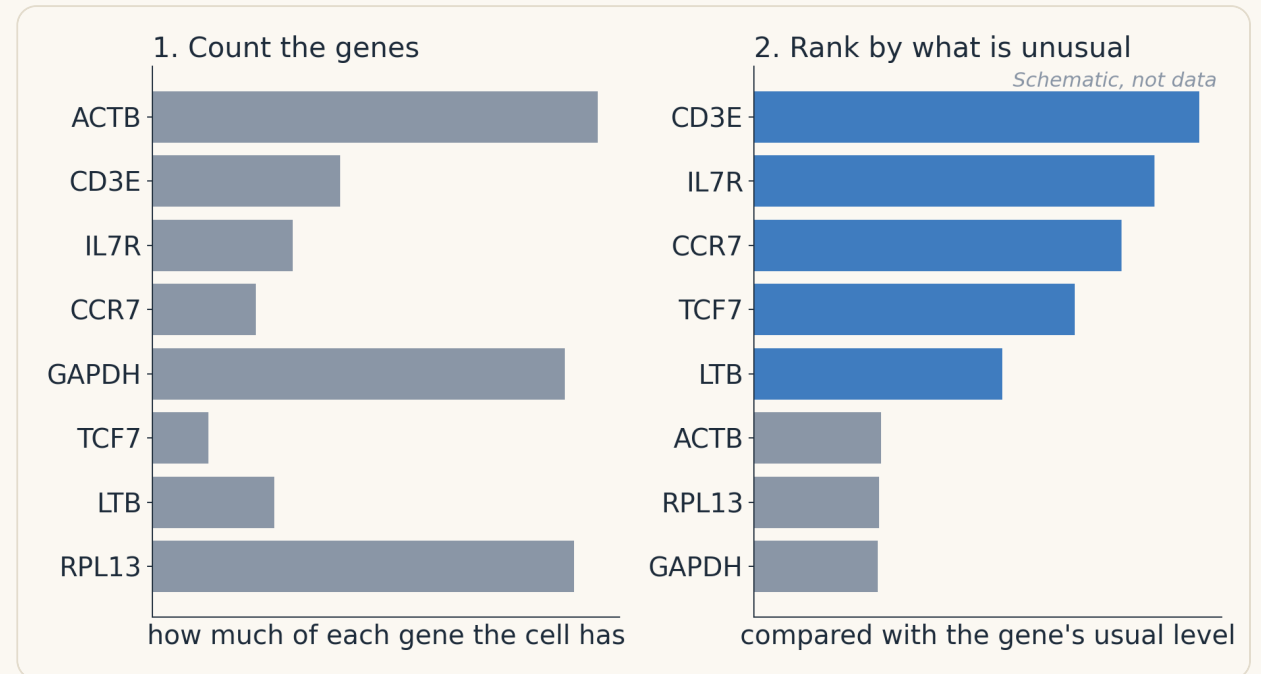
1. **The model.** What Geneformer is, in plain terms.
2. **Editing a cell.** How a "virtual gene edit" works and how we score it.
3. **How it fooled us.** Four moments, July to September, when a result looked real and was not, or not entirely.
4. **Checks.** Six rules we now apply, each born from one of those moments.
5. **What we know.** What our results support today, and what they do not.
6. **Next.** What we still need to test, and the paper it could become.

Colour code throughout: ✓ passed ✗ failed ? not yet known

Geneformer reads each cell as a list of its genes, most unusual first

A cell contains thousands of genes at different levels. Geneformer compares each gene with its usual level across many cells, then sorts. Genes that are always high (such as ACTB) sink; genes that make this cell distinctive rise to the top.

The sorted list is the cell's "sentence". Each gene in it is a **token**, the model's word.



It learned what normal gene lists look like from about 104 million cells



Training game: hide a gene, ask the model to guess it from the rest of the list. Illustration, not data.

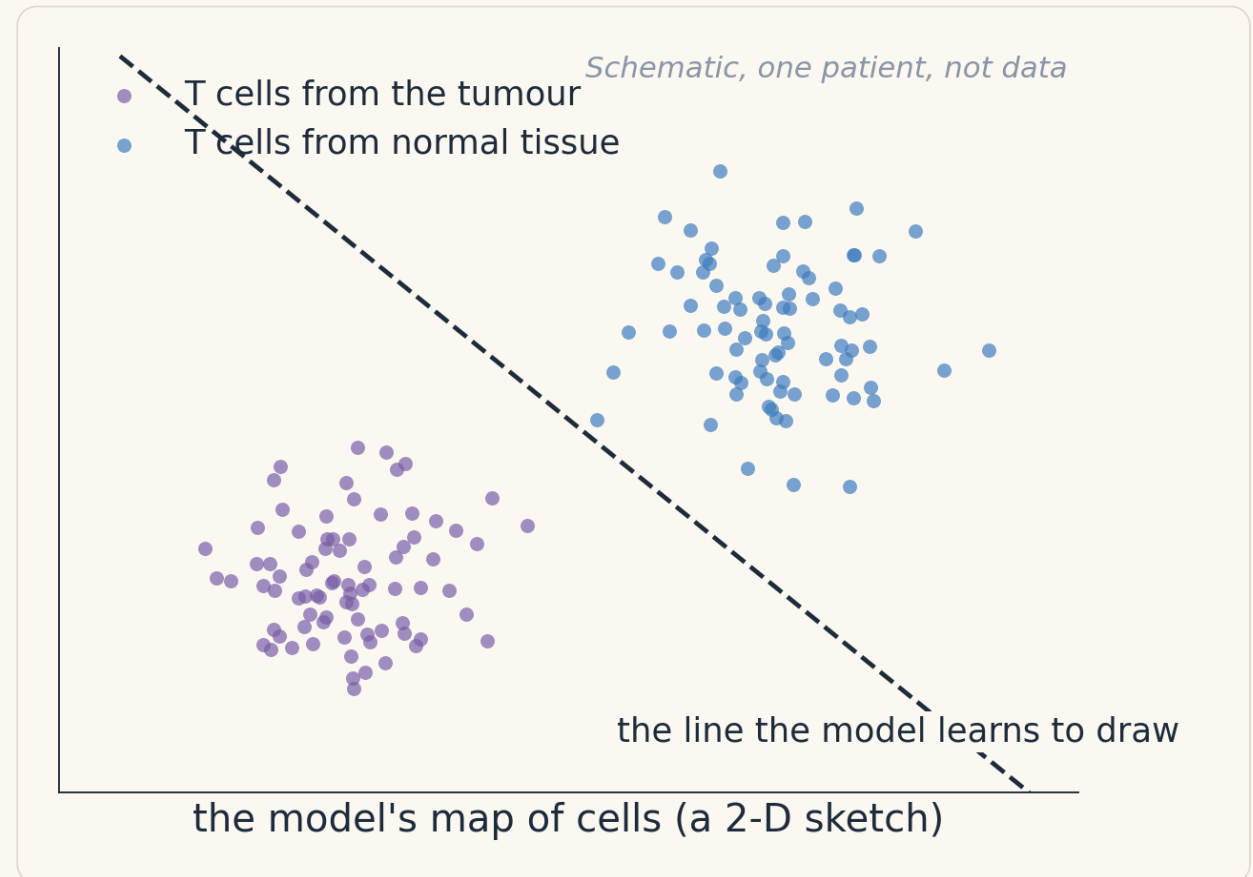
After playing this game on about 104 million human cells, the model places every cell at a point on an internal map, its **embedding**. Cells with similar lists land close together.

The model comes in sizes; we have used a smaller and a larger one.

We then teach it one extra skill: telling a patient's tumour T cells from their normal T cells

This extra training is called **fine-tuning**. We always test it on patients the model has never seen.

It worked well in lung cancer: right **82.5%** of the time, counting tumour and normal cells equally (this measure is called balanced accuracy; a coin toss scores 50%), and better than a coin toss for all **43 of 43** patients.

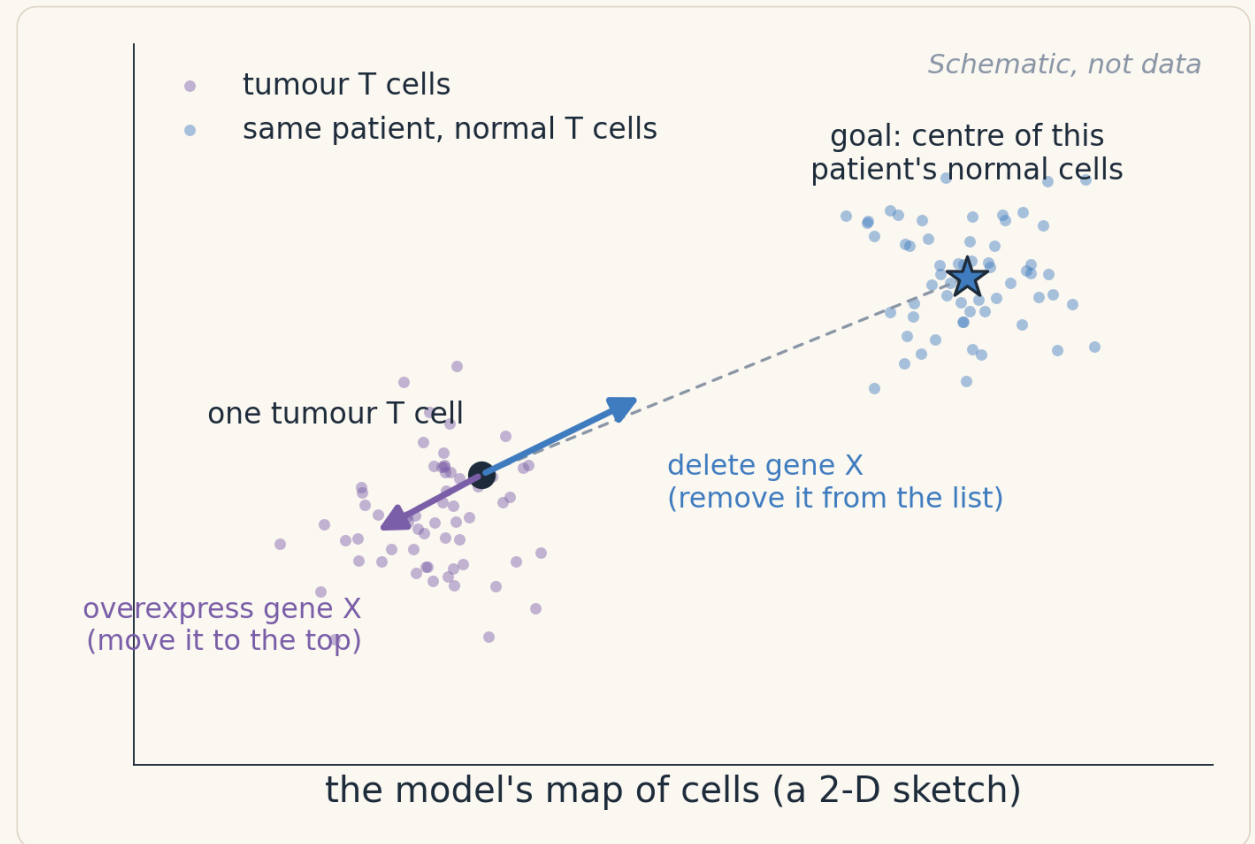


A virtual gene edit changes the list and asks where the cell moves on the map

Delete a gene: remove it from the cell's list. **Overexpress** it: move it to the top.

Then measure whether the edited cell moved **toward the centre of that patient's normal T cells**. That movement is our score.

This is called in-silico perturbation (ISP). Nothing happens to a real cell; it is a question put to the model.



Our first results, in July and August, looked encouraging

July: a whole-genome screen

21,000 lung T cells, the smaller model, and almost 3 million single-gene deletions (2,937,776).

Our own audit, 28 July: the lung adenocarcinoma and squamous-cell work could not on its own support what our conference abstract claimed about small-cell lung cancer.

August: a conference talk

We tested 50 genes in small-cell lung cancer (SCLC). Four passed in both directions in all three test patients: deleting moved cells toward normal, overexpressing moved them away.

They were TIM-3, TIGIT, CTLA-4 and IL7R, genes linked to T-cell exhaustion or persistence.

Trap 1: the strongest hits were not T-cell signals

Top of the screen

S100A8 and S100A9, among our most significant genes, moved cells the *same* way whether deleted or overexpressed in 9 of 12 comparisons. A real driver should not.

Of the top 120 results, 55 were flagged as likely **ambient RNA**: traces of other cells' genes picked up during sample preparation.

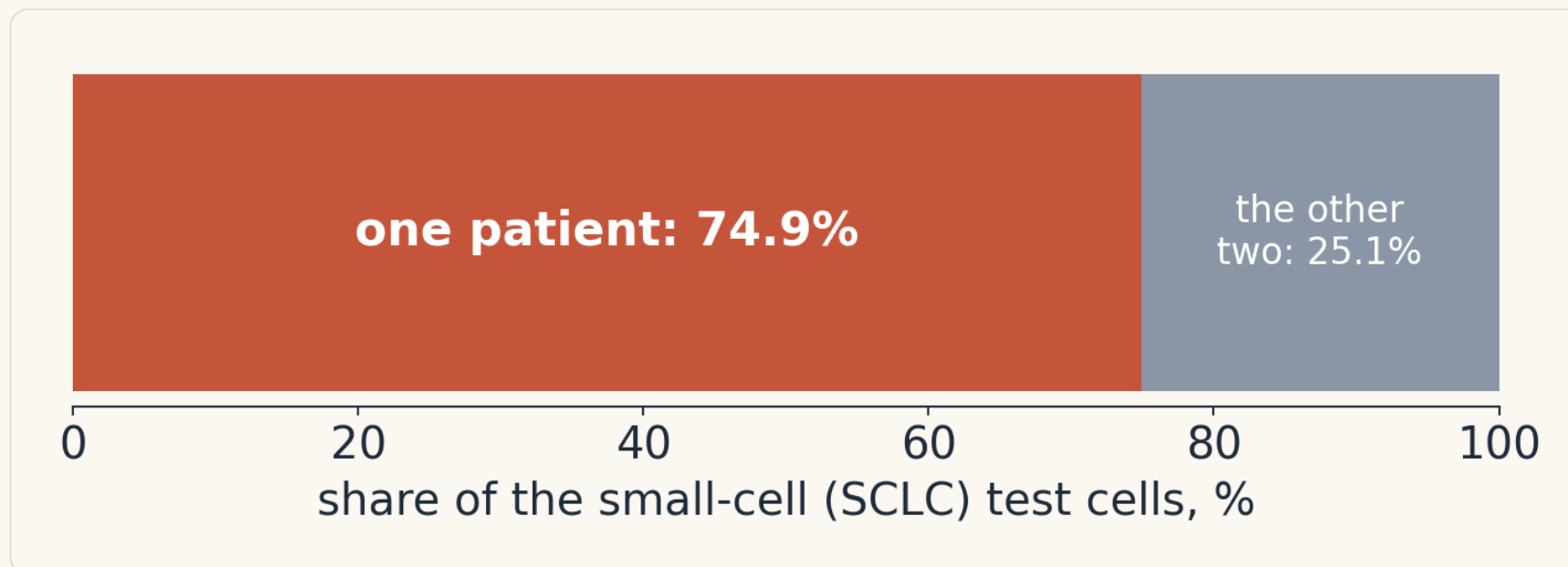
The "normal" group

In July, the normal and tumour T cells came from different studies: none of the 18 studies supplied both. The model could partly have learned "which lab" instead of "which tissue".

Later, 13 of the 15 July top genes could not be tested in T cells at all.

Answered by  2  4

Trap 2: one patient can look like a whole group



Two of our analyses seemed to disagree about small-cell lung cancer. The cause was one patient who supplied three quarters (74.9%) of the small-cell test cells. Counting patients instead of cells (19 against 22 patients), the difference was not significant: $p = 0.216$.

Answered by  6

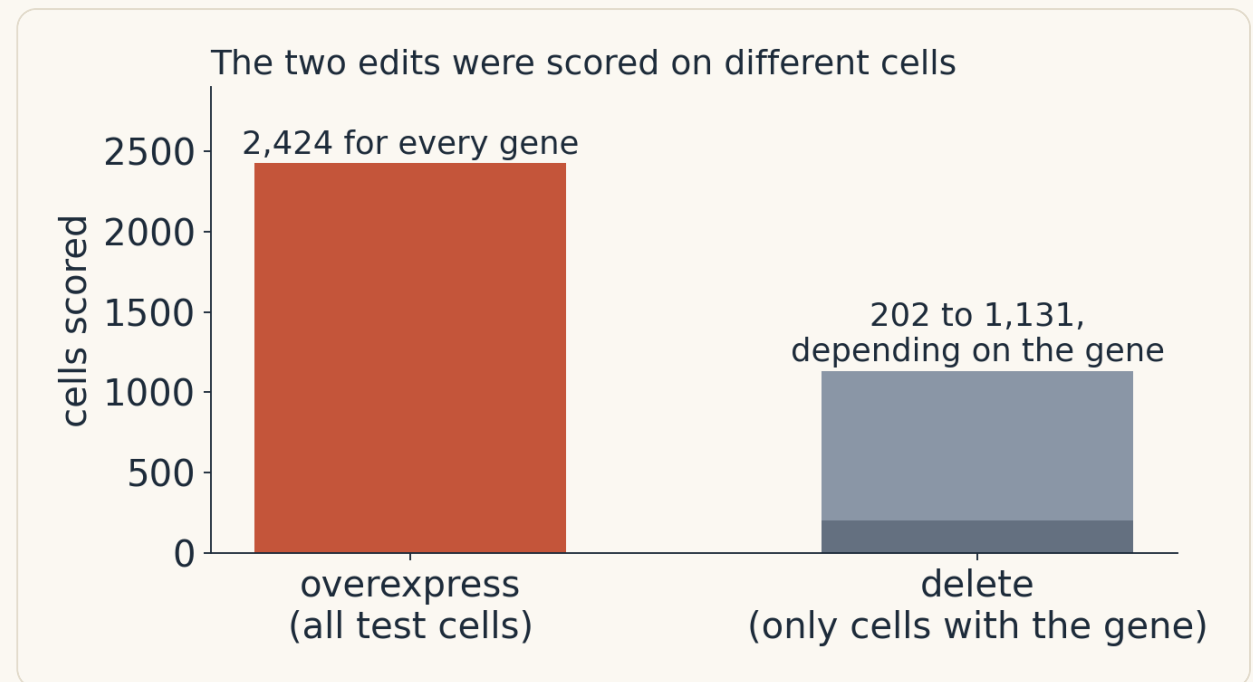
Trap 3: our two edits were measured on different cells

In the August panel, overexpression was scored on every test cell, but deletion only on cells that already carried the gene. Comparing the two arms compared different cells.

The cause was a default in Geneformer's code: given a list of genes, it inserts an overexpressed gene into every cell (14,738 of 15,179 calls in our rebuilt study).

We attached a caveat to the four August genes: "a qualifier, not a retraction".

Answered by  3



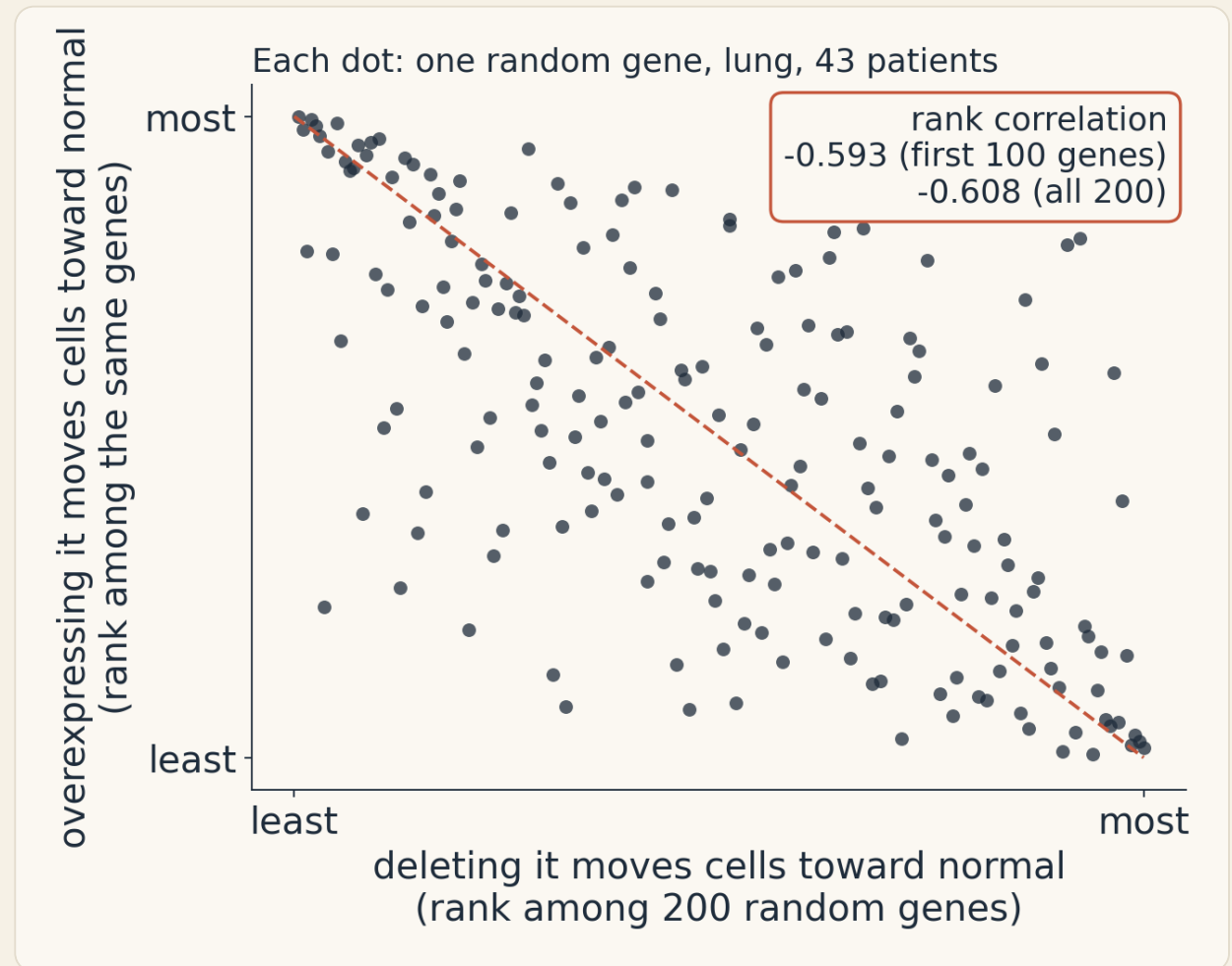
Trap 4: even random genes show the pattern we had treated as evidence

Opposite effects of the two edits looked meaningful. So we ran the same pipeline on **random genes** with no known role.

They did it too: rank correlation **-0.593** for 100 genes and **-0.608** for 200 (-1 is perfectly opposed, 0 unrelated).

Chance alone would give this at most about once in 100,000 tries, the smallest value our test can report.

Answered by  5



Six checks now stand between a result and a claim



1

Write the test down first

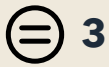
fix the test, the threshold and the wording before any result



2

Check the instrument

the model must work on unseen patients; an empty edit must give zero



3

Same cells for both edits

delete and overexpress are scored on identical cells



4

Beat look-alike genes

a gene must move cells more than genes detected as often



5

Beat random genes

a pattern random genes also show is not evidence



6

Count patients, use fixed words

patients are the unit; results must be stable; outcomes get fixed names

Written down on 28 September 2026 as the lab's standard for judging virtual gene edits (ISP-STD-1).

1 Write the test down before any result exists

Before running anything, we record the test, the direction we predict, the threshold for calling it significant, and the words we will use for each outcome. An independent reviewer signs off on the exact files.

We also check that the test **can** reach significance with the data we will have. Small studies sometimes cannot.

Learned from: August, when we read meaning into an ordering we had not planned to test.

EXAMPLE: THE COLON STUDY

We expect 10 lung reference genes to be testable in colon. To call the lung pattern repeated, at least **9 of 10** must keep their lung direction. (If only 9 can be tested, 8 of 9.)

The first review of that plan failed: it claimed the software was unchanged, but one library had been downgraded. We corrected it before starting.

2 Check the instrument before reading anything from it

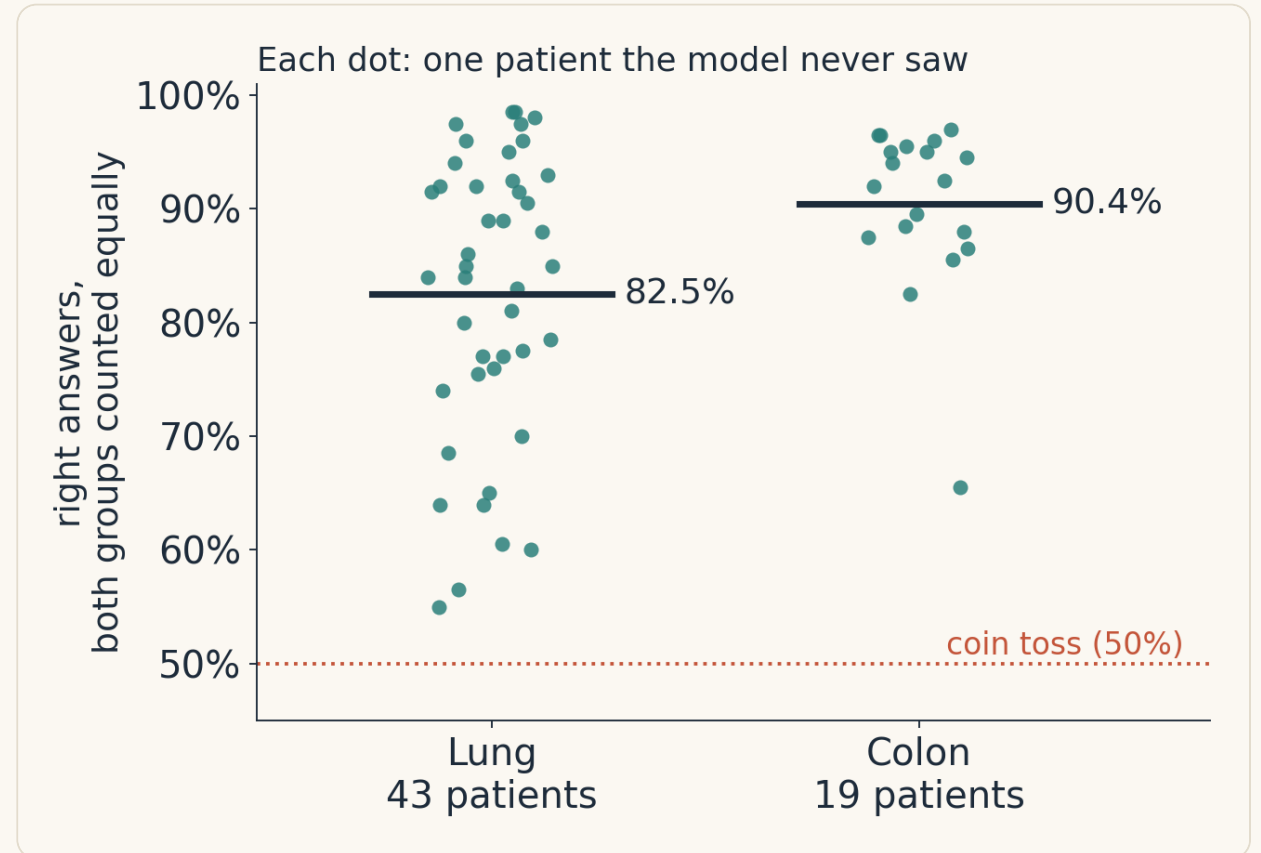
Does the model work? It must tell tumour from normal T cells in patients it never saw, better than chance across patients. If not, we stop.

Does it stay still? Run the pipeline twice on identical input with no edit. Any movement is noise that would later look like an effect.

Colon, this week: ✓ 90.4% right; 19 of 19 patients

✓ empty edit moved 0.0

Learned from: the July model, which may have been telling studies apart.



⊖ 3 Score both edits on exactly the same cells

✗ August panel

Overexpress: 2,424 cells for every gene.
Delete: 202 to 1,131 cells.

✓ Since 25 September

Both edits are analysed on the same cells, and every run is checked for it. (Panel code fixed 22 September; rebuilt study corrected 25 September.)

A simple warning sign: if one edit always uses the same number of cells while the other varies by gene, the arms are not paired.

| Learned from: trap 3.

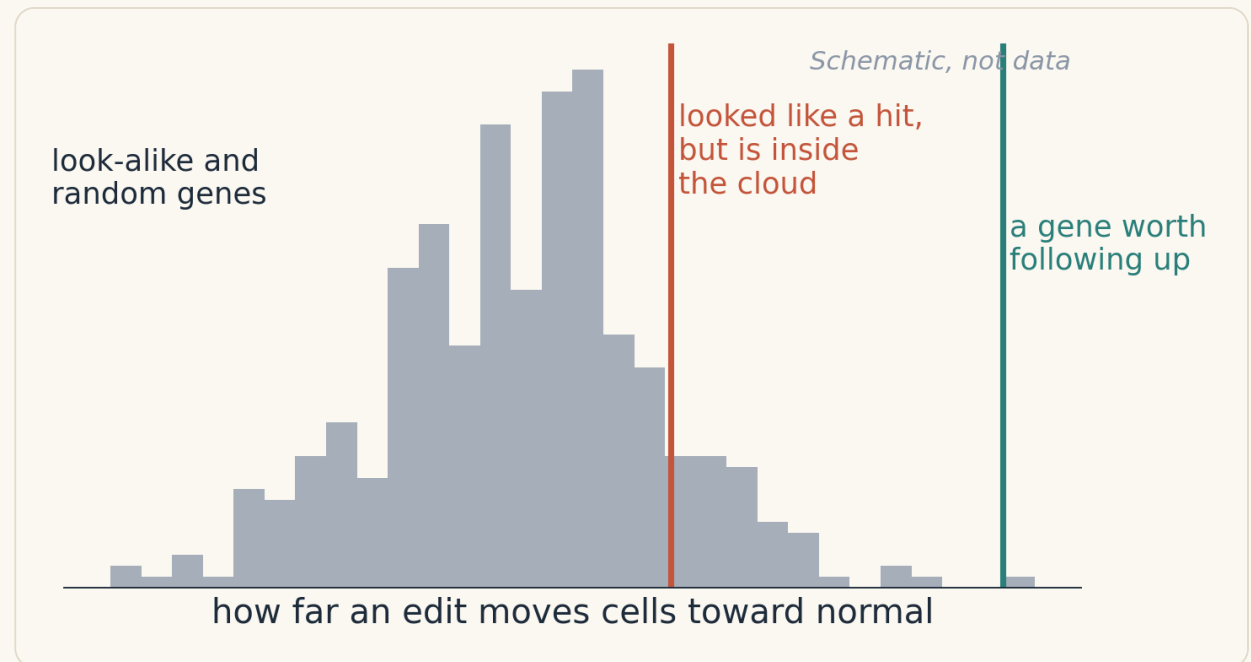
⚖️ 4 A gene must move cells more than genes that look like it

Genes found in many cells move cells more, whatever they do. So each gene is compared with **at least 20 look-alike genes**, detected about as often and ranked about as high.

If 20 cannot be found, the gene is reported as *not estimable*. We never loosen the match to get an answer.

Genes likely to be ambient RNA, or used to build the classifier itself, are flagged.

| Learned from: trap 1 (S100A8, ambient RNA).



5 Opposite effects of delete and overexpress are not, on their own, evidence

What the random genes show

Opposed effects are what this model and this design produce for genes in general. They are the baseline.

The study's own control genes agreed: rank correlation -0.658 (308 of 318 genes had enough data).

What they do not show

They do not erase the panel result: 13 of 34 testable T-cell genes still beat their look-alikes.

They change what counts as evidence: an unusually *large* effect, measured against the random-gene baseline, not opposite signs.

| Learned from: trap 4.

☹️ 6 Count patients, demand stable results, and name every outcome in advance

Patients and stability

The patient, not the cell, is the unit of evidence.

A result must survive dropping any single control gene, and hold in at least 95% of 10,000 resamplings.

Fixed words

positive

negative

opposite_direction

not_estimable

open

We never write "no effect". A negative result means this test, at this size, did not meet its bar.

Learned from: trap 2 (one patient) and the August wording.

What our results support today, and what they do not

✓ Supported (lung, 43 patients)

- A fine-tuned model separates tumour from normal T cells in unseen patients (82.5%; 43 of 43 above chance).
- 13 of 34 testable T-cell genes move cells more than their look-alikes, stably.
- Delete and overexpress give opposed effects for random genes too.
- Colon: the tumour/normal split is learnable too (90.4%; 19 of 19).
- Colon: the lung gene pattern did not repeat (3 of 10 genes kept their lung direction).

? Not shown

- That any gene controls T-cell state. A move on the model's map is not a mechanism.
- Whether the opposed effects come from the model or from lung tissue: in colon they point the same way but are not stable (next slide).
- What the classifier actually reads: T-cell state, ambient RNA, or the mix of T-cell types.
- Whether the lung models work on new lung patients.

The colon test: random genes were opposed again, but not reliably; the lung gene pattern did not repeat

? Open Random genes

Delete and overexpress again pushed random genes in opposite directions: rank correlation -0.25 (lung: -0.59), $p = 0.0067$.

It survived dropping any one gene, but held in only 78% of resamplings; we required 95%.

Our pre-written reading: same direction as lung, not stable, so the question stays open.

X Not repeated Lung reference genes

Of 10 lung genes we could test, 3 kept their lung direction; we required 9. Seven pointing the other way is within what chance gives.

One gene, PRF1, passed every check in colon; in lung it was undecided.

We cannot blame tissue for the differences: the colon data come from another study, with different sample handling, new models and fewer patients (19 against 43).

Summary 1: the evaluation criteria we found, each traced to evidence

CRITERION	✓ ADOPTED	EVIDENCE THAT TAUGHT IT
 1	Register test, direction, threshold, reachable p and wording first	August ordering read without a plan
 2	Model must work on unseen, paired patients; empty edit gives zero	July groups came from different studies (0 of 18)
 3	Both edits on identical cells	2,424 vs 202 to 1,131 cells; code default 14,738 of 15,179
 4	Beat 20 or more look-alike genes; flag ambient genes	S100A8 and S100A9: same direction in 9 of 12; 55 of top 120 ambient
 5	Beat random genes; opposite signs are not evidence	Random genes: −0.593 (100), −0.608 (200)
 6	Count patients; stable in 95% of resamplings; fixed outcome names	One patient held 74.9% of the SCLC test cells

Status: adopted as lab practice (ISP-STD-1, 28 September 2026). Established as practice, not yet published or tested by others.

Summary 2: what we still need to test

STATUS	QUESTION	HOW
? open	Are the opposed effects a property of the model or of lung?	Colon (E2): same direction, not stable; lung gene pattern did not repeat
? blocked	Do the lung models work on new lung patients?	E1: the one candidate dataset is not yet accessible
? planned	Are the opposed effects the same whatever the goal? (calibrating the random-gene baseline)	Swap the goal for an unrelated one
? planned	Is the 13-gene result cell state, or just a different mix of T-cell types?	Re-analyse within T-cell subsets
? planned	Can a simple, readable method match the model's 82.5%?	T-cell program scores (TCAT, Kotliar et al. 2025)
? not started	Do results change with model size (104M vs 316M parameters)?	Re-run a subset on both sizes
? not started	Does the method find genes we already know matter?	Positive controls; real gene-editing screens in T cells

All rows are open. The colon row has a first answer: open for random genes, not repeated for the lung genes.

Together these could become one paper on how to judge virtual gene edits

Working title: "Opposite by default: baselines that in-silico perturbation in single-cell foundation models needs"

Result. ✓ established in lung, larger model, our design Random genes give opposed delete and overexpress effects (rank correlation -0.59 and -0.61).

Our argument. Gene-level claims therefore need look-alike and random-gene baselines; the traps add paired cells, patient-level counting and a plan filed first.

? hypothesis The same holds in other tissues, for other goals and model sizes.

Fig. 1 Design and the six checks

Fig. 2 Random-gene baseline in lung

Fig. 3 The same in colon, and under a swapped goal

Fig. 4 Four traps the checks catch

Fig. 5 T-cell genes against look-alikes

Fig. 6 Model size, new lung patients, known genes

The model always answers. The checks tell us when to listen.

Measure what random genes do first, then claim what one gene does.

Every number is sourced in SOURCES.md in this talk's folder.

Glossary: the few terms we kept

Geneformer	A model trained on about 104 million cells, each written as a ranked list of genes.
Token	One gene in a cell's list: the model's word.
Embedding	A cell's position on the model's internal map; similar cells sit close together.
Fine-tuning	Extra training on one task, here tumour versus normal T cells.
In-silico perturbation (ISP)	A virtual gene edit: delete a gene from the list or move it to the top, then measure where the cell moves.
Balanced accuracy	How often the model is right, counting both groups equally; 50% is a coin toss.
Ambient RNA	Genetic material from other cells picked up during sample preparation.
Look-alike (matched control) genes	Genes detected about as often and ranked about as high as the gene being tested.
Rank correlation (ρ)	From -1 (perfectly opposed) through 0 (unrelated) to 1 (perfectly aligned).
p-value (p)	How often chance alone would give a result at least this strong; small means unlikely by chance.
GPU-hour	One hour of work on one graphics processor, the chip that runs the model.
Pre-registration	Writing down the test, its threshold and its wording before any result exists.